Imfroment in Food Resource

- Food:- Any Organic substence that contains nutrients. i.e; Proteins, Carbohydrate, Fats, Vitamins, Minerals.

- Consumed by living organisms in order to derive energy and support all the vital metabolic activities.

- Food Resource:-(Crop)

- A crop is a cultivated population of same type of plants, grown and managed over a larger land area.

1) Food Variety Improvement:-

- Focuses on selection of plants with desirable qualities.
  ↳ Higher Productivity → Incrased production per acre.

→ Improved quality = Pulses → Protein
= Oil seeds → Oil quality.

→ Increased shelf-life = Fruits and vegetables.

⇒ **Biotic and Abiotic resistance :-**

- Crop production can go down due to certain biotic factor like - Insects, diseases, nematodes.

→ ~~Ab~~ Abiotic factors like - drought, salinity, water lodging, heat, cold, frost.

⇒ **Change in maturity duration.**

- ~~Shorte~~ Shorter duration of crops from sowing to harvesting can help farmers to grow multiple rounds of crops.

⇒ **Wider adaptability.** Developing various adaptations pertaining to environment i.e; climate conditions, weather.

⇒ **Desirable agronomic trait/character:**

- Fooder → plant → Tall.
- Food crop → dwarf (short) → because the nutrients utilized for height, will be instead used for production of grains/fruits.

2) Crop Production Management:

- It refers to controlling, organizing and optimizing various practices to ensure maximum yeild.
- It is how a farmer manages, nutrients water supply and cropping techniques to help healthy and

abundant growth of crops.

→ **Nutrient Management**-

- There are various types of nutrients required by plant.

⇒ Among these are the nutrients required in large quantities, these are known as - macronutrient

e.g Carbon, hydrogen, oxygen, ~~the~~ (CHO). (NPK) → Nitrogen ~~phos~~ Phosphorus Potasium.

⇒ While as the nutrients required in lesser amount are known as micro-nutrients.

e.g Copper, zinc, Nickle

Nutrient

↓ Manure ↓ Fertizer

⇒ **Manure**:- It is a natural organic substance, produced by decomposition.

- Substences like plants and animals waste.

⇒ **Types of Manure**:-

1) **Compost**:- It is prepared by decomposing agricultural waste, household wastes (Vegetable peel) and live stock wastes.

2) **Vermi-Compost**:- Compost prepared by using earth worm.

⇒ **Green Manure**:- The Fast-growing plants are grown and then ploughed and mulched with soil. e.g.- Sun-hemp.

⇒ **Benefits**:- It provides
- Nutrients
- Increase water holding capacity.
- Improves texture of soil and aeration
- Reduces reliance on synthetic chemical which causes soil polution.

⇒ Fertilizer:

- These are commercially produced synthetic substances or chemical compound.

e.g. NPK fertilizer.
Urea fertilizer.

→ Easy to absorb.

→ Target essential nutrient requirement.

⇒ However excessive use of fertilizers cause harm to soil microbes.

→ Change PH of soil.

→ Washing out of these nutrients into water bodies cause eutrophication.
→ Nutrient enrichment of water bodies.
→ Growth/excessive growth of algae.
→ Depletion of water oxygen i.e; dissolved

oxygen.

→ **P.H.:-** PH means Potential of hydrogen ion.

- A scale used to measure, how acidic, basic a substance is.

→ **Decomposition:-** Break down of substance, by microbes/micro-organism.

⇒ **Irrigation/Water Supply:-**

- Indian agriculture depends on moonsoon for water supply.
- But the crops require water growing period.
- Different method for Water supply.

1) Well. (2) Cannal. (3) River/riverlift system.
4) Tanks.

1) Well.

Dug Well.	Tube Well.
• This well draws water from bearing strata/foundation strata.	• It draws water from deeper strata/layer. • Water is uplifted with the help of the pump.

2) Cannal-

- Extensive and elaborate water supply system.
- Cannal divides into branches.
- And branches into field.

3) River Lift System:-

- When Cannal has inadequate supply of water then the water is directly drawn from rivers.

4) Tank:- Small storages, that collect run-off

water from catchement areas.

→ Crop Pattern -

- Different ways of growing crops, in order to get maximum benefit and yeild.
- It is of following types:

a) Mixed Cropping

- Grow two or more crops on same piece of land without following any specific pattern.

- Benefit - This method reduces risk or acts like Insurance aganist complete crop failure.

e.g. Wheat + Gram
Wheat + Mustard

b) Inter - Cropping - Growing two or more crops together on same land but following a specific pattern of rows.

- Benefit- This method ensures maximum utilization of nutrients.

- It prevents ~~speed~~ spread of disease to all the crop plants.

e.g Maize + Soyabean.

Cow pea (~~bajra~~ lobia) + finger millet (~~lobia~~ bajra).

→ Crop rotation-

- Growing different crops on same land but in pre-planned succession year after year.

→ Benefit- Growing different crops yearly ensures fertility of that particular ~~year~~ land area.

Page No.
Date / /20

⇒ Crop production Management -

- It refers to all agricultural practices and complete set of decision by a farmer, during cultivation and storage after harvesting.

⇒ Crops can be damaged by:

1) Weeds
2) Insects
3) Diseases.

1) Weed - It is an unwanted plant.

- Impact - It competes with :- nutrients, space, sunlight
  → over all growth of ~~of~~ crops is reduced.
  e.g Parthenium (Gajar Grass).

2) Insects - They damage crops by three method.
  1) By cutting root, stem and leaves.
  2) Draw sap from different parts of plant.
  3) Boring into stem and ~~leaves~~ fruit.

3) <u>Plant Diseases</u>- A plant gets diseased by bacteria, Viruse, Fungi.

- These diseases spread through, soil, water, air.

⇒ <u>Preventive ~~ch~~ and control ~~R~~ measures</u>:-

a) <u>Chemical method</u>-

Spraying of pesticides
- Herbicide.
- Insecticide.
- Fungicide.

<u>Pest</u>- Any Animal/Insect that damages crop.
<u>Cide</u>- To kill.

(b) <u>Mechanical or cultural method</u>:-

- Removal of Weeds by hand.
- Proper seed bed making.
- Timely sowing.
- Crop rotation
- Inter cropping.

c) Preventive measure -

- Use of disease resistant varities.
- Summer ploughing.

(4) ☆⇒ Storage of Grains -

- High storage ~~los~~ losses can drastically reduce the ~~production~~ produce quality and marketability.

⇒ It can occur by following factors.

1) Biotic factors :- Bacteria, fungi, mites, rodents.

2) Abiotic factors : Inappropriate moisture content and temperature during storage.

⇒ Effects of storage degradation :-

- Quality degradation.

- Weight loss.
- Discolouration
- Poor germination
- Low market value.

⇒ Prevention and control measures:-

- Strict cleaning:- Thoroughly cleaning of grains before storage.

- Drying:- First sundrying, follow by shade.

- Fumigation:- Using safe chemical to kill pests.

  - It is a pest control method in which sealed space is filled with ~~ga~~ gaseous chemicals, in order to eliminate pests.

Seen
31/9/26

⇒ Animal husbandary:- (Live Stock).

- It is the ~~of~~ scientific management of animal live stock (feeding, breeding, disease control).

- ~~Dr~~ It includes = ~~Crop~~ Cow.
  = Goat.
  = sheep
  = Poultry
  = Fishes
  = Honey bees.

⇒ **Need**- Due to increasing population.

- Rising living standards.
- Increased demand for meat, milk and eggs.

⇒ Cattle Farming:- Two Reasons/Purpose:-
1) Milk → Milch animals
2) Agricultural purposes → Drought Animals.

★⇒ Lactation:- The period of milk production after birth of young one (calf)

☆→ Cow :- Bos Indicus. (scientific Name).
☆→ Buffaloe :- Bos bubalis. (scientific Name).

⇒ Indigenous Varities of Cow :-

- Red Sindhi
- Sahiwal.

⇒ These varities have disease resistance.

⇒ Exotic Varities :- High milk production.

- Jerrsey
- Brown Swiss

⇒ Cross breading.

- Breeding done between exotic and indigenous varities, in order to get both desirable qualities i.e ~~dis~~ disease ~~and~~ resistance and high milk yeild.

⇒ Shelter Management:

- Animal shed needs to properly cleaned regularly.
- Animals need to be regularly brushed, so that extra loose hairs and dirt is removed.
- Their shed should have proper ventilation, and roofed so that to protect animals from, cold, heat and rain.
- The floor of shed should be sloping, so as ~~it~~ to stay dry and facilitate cleaning.

⇒ Food requirements:-

→ Food requirements are of two types:

i) Maintenance requirement: The food required to maintain healthy life.

ii). Milk Producing requirement:- The food required during lactation period.

⇒ Animal Feeds Include:-

- Roughage = Contain fiber.
- Concentrates:- Low fiber but high nutritional value.
- Feed Additive:- Contains food that can promote health and milk production.

⇒ Disease Management of Cattles:

- Animals get disease by number of pathogens (disease causing organism.) like virus, bacteria, fungi.
- Animals can be vaccinated to prevent diseases.

⇒ **Poultry Farming:-**

- The Practise done to raise domestic fowl for eggs/Chicken meat.

⇒ Eggs:→ Fowl is known as layer.*
⇒ Chicken Meat:- Fowl is known as Broiler.*

⇒ Different cross-breeding programmes have been done between Indian indigenous variety - ~~Asel~~ Aseel* and exotic/foreign variety → laghorn* to create better varities.

⇒ Following desirable qualities:-

- Number and quality of chicks.
- Dwarf broiler parent for commercial chicken production.

- Temperature Tolerance.
- Low maintenance requirements.
- Reduction in size of egg lying bird with ability of utization of cheaper fibrous feed. produced from agricultural by-product.

⇒ <u>Maintenance of Layers and broilers:</u>

- Chicken should be fed with vitamin rich food for:
- Good growth rate
- Avoid mortality. (death).
- Maintain feathering.
- Caracus quality
  ↳ body of slaughtered chicken.

⇒ Good management practices also Include:

- Hygienic condition
- Maintenance of Temperature.
- Proper feed.
- Prevention of diseases.

☆⇒ Poultry suffers from nutritional deficiencies and diseases.

↳ Broilers have different nutritional requirement compared to layer.
- So their diet should contain protein, adequate fat and high levels of vitamin A and K.

- Proper cleaning and sanitation and spraying disinfectants.
- Appropriate Vaccination.

☆⇒ Fish Production:- (Pisciculture).

- The Practice of rearing, breeding and managing fishes for commercial production.

- It serves as major source, low-cost animal protein

⇒ Methods of obtaining Fishes:

1)→ Capture fishing :- Obtaining fishes directly from ~~no~~ natural source. e.g Ocean, Sea, river, lake, etc.

2) Culture Fishing :- Cultivating and breeding the fish in controlled water body.

Good

Seen
9/1/20

⇒ Types of Fish culture based on type Water.

Marine culture.
↓
Culture done in saline Water.
→ Either at shore.
→ OR away from shore/deep sea.

Inland culture.
↓
Culcute done in fresh water.

⇒ Popular varities in ~~in~~ Marine :- = Pomfret
= Meckerol

= ~~Srd~~ Sadines.
= Tuna
= Bombay dock.

⇒ Popular Varities (Inland Culture):- = Rohu
= Catla.
= Grass carp
= Silver carp
= Common carp.

★⇒ Some economically Important culture Varities:-

- Finned Fishes:- Mullets, Bhetki, Pearl spot.
- Shell Fishes. Prawn, mussels, oysters.

★- Oysters are cultured for pearl Production.

⇒ Important Terminologies:-

- Mariculture:- Culture of marine organisms
- Aquaculture:- Culture of fresh water organisms.
- Integrated fish culture:- Culturing of fishs with

crop farming.

e.g Fishes in rice paddy field.

⇒ <u>Composite fish culture/polyculture:</u>

- Growing 4 to 5 fishes together.

⇒ ~~Surface~~ <u>Feeding habit distribution:</u>

Surface feeders	Column feeders	Bottom Feeders
There fishes feed on surface. e.g Silver carp catla.	These fishes feed in middle zone. e.g Grass Carp Rohu.	The fishes feed at Bottom of water body. e.g Common Carp mrigal.

⇒ <u>Advantages:</u>

- It reduces competition.

- Maximum food utilization

- Higher yeild.